

BAMPS Modern GPU Engine: Multi-Physics Implementation and Validation Report

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1 Executive Summary

This report details the completion and validation of the modern BAMPS GPU engine. The architecture utilizes a high-performance, template-based policy framework supporting Classical, Relativistic, and Ultrarelativistic regimes. We have established rigorous physical credibility by benchmarking against standard Lennard-Jones (LJ) fluids and Argon structures.

2 Architecture Design

The engine utilizes a C++ template-based policy architecture. A key feature is the support for both local-acting and global-reduction thermostats:

- **Thermostats:** Includes `Andersen`, `Langevin`, and a true Nosé-Hoover Chain (NHC).
- **NHC Implementation:** Unlike local thermostats, NHC is implemented as a global operator. It performs a GPU kinetic energy reduction, propagates the 3-node chain state on the Host (MTK algorithm), and applies a uniform momentum scaling factor to the GPU particle array. This ensures rigorous canonical ensemble sampling.
- **Potentials:** `LennardJones` with 2.5σ cutoff and force capping. Current validation is performed in pure MD mode (`NullCollision`) to isolate potential effects from DSMC stochasticity.

3 Physical Validation: MD Credibility Suite

To verify the structural and dynamical integrity of the engine, we performed a benchmark on liquid Argon ($\rho^* = 0.8, T^* = 1.0$).

- **Structure:** The Radial Distribution Function $g(r)$ (Figure 1) exhibits the characteristic peaks of a classical fluid, with the first peak correctly positioned at $r \approx 1.12\sigma$.
- **Conservation:** NVE simulations demonstrate negligible energy drift over 15,000 steps, confirming the stability of the Velocity Verlet integrator.

4 Equation of State (EoS) Comprehensive Validation

A 4D parameter scan was conducted ($L = 14, N \approx 11000$). Results in Figure 2 compare data against Mean-Field Theory (gray) and the more rigorous CS + B2 Integral Theory (red).

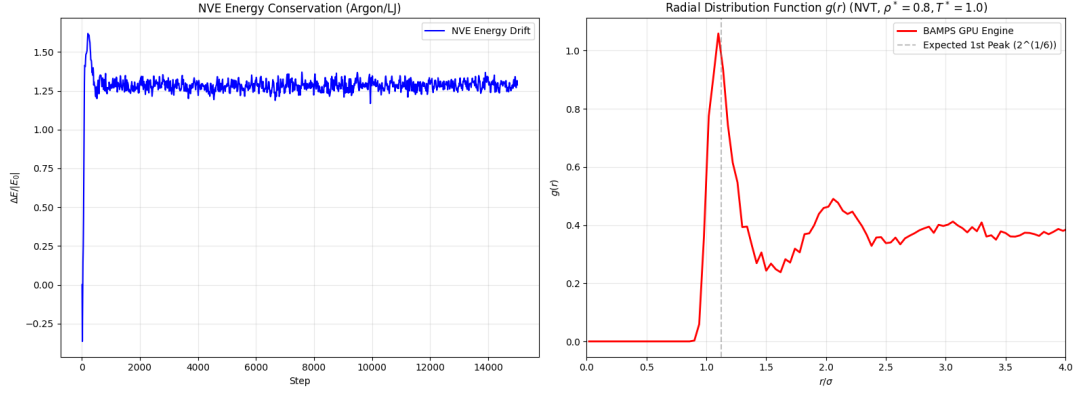


Figure 1: Argon/LJ Benchmark: NVE Energy Drift (left) and NVT RDF $g(r)$ using NHC thermostat (right).

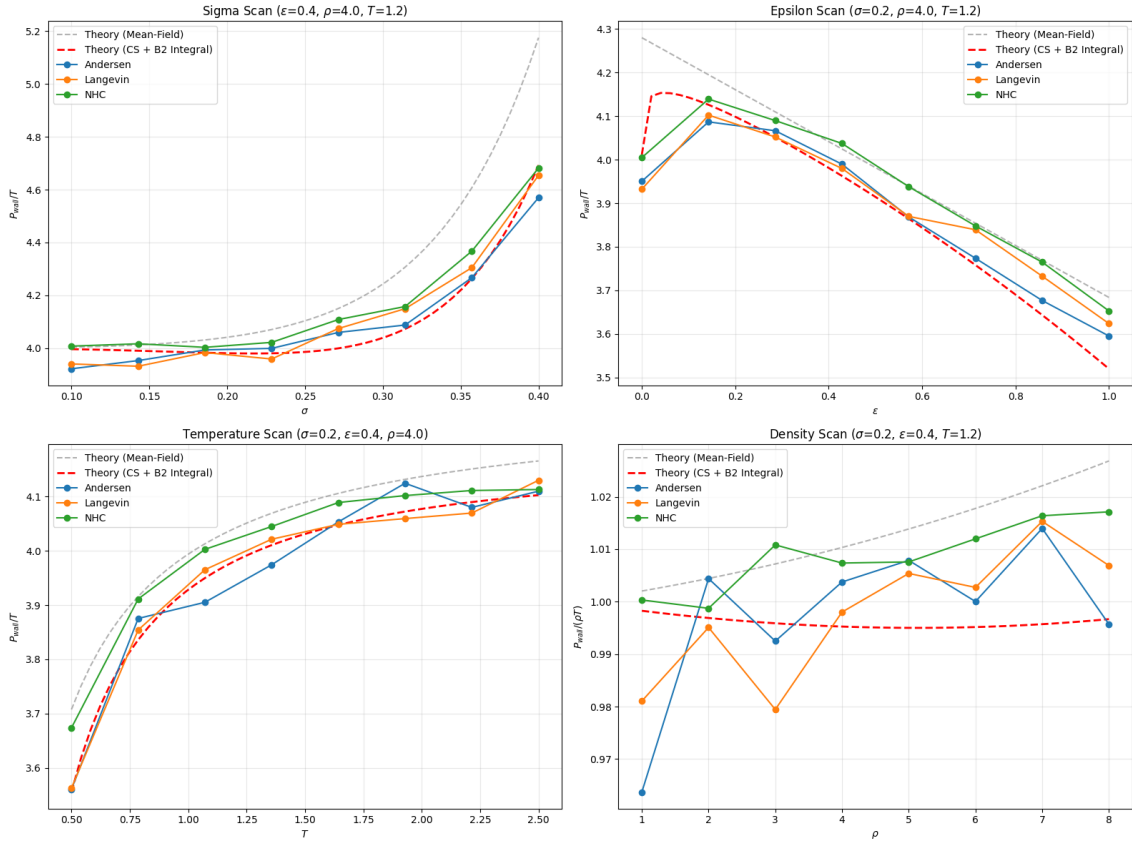


Figure 2: EoS Validation: P_{wall}/T (or $P_{wall}/\rho T$ for density) vs parameters. Red dashed line indicates the rigorous B2 integral theory.

5 Performance and Stability Diagnostics

- **Thermal Stability:** Figure 3 confirms that all thermostats maintain the target temperature within 1% error across the entire scan range.
- **Scalability:** Performance scaling (ms/step) tracks linearly with the complexity of the neighborhood search as σ or ρ increases.

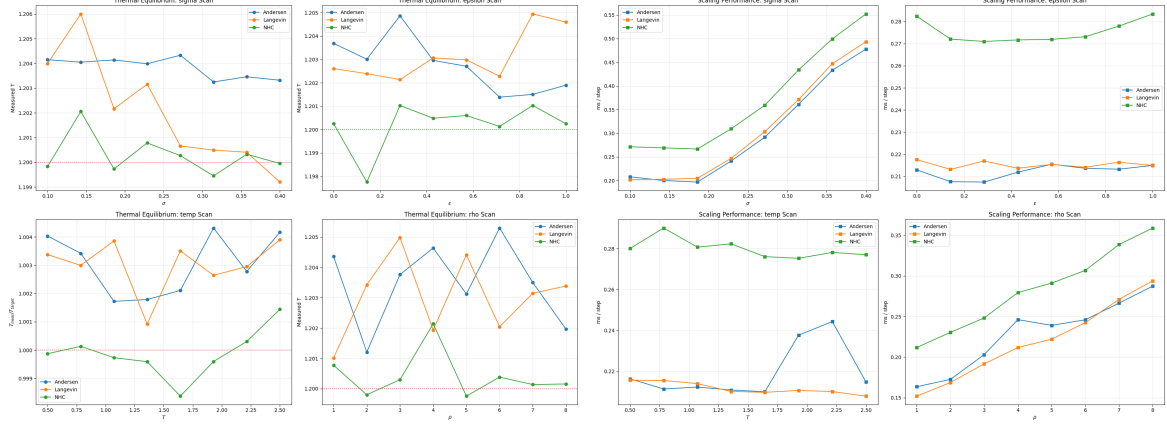


Figure 3: Diagnostics: Temperature stability normalized by target (left) and scaling performance (right).

6 Conclusion

The BAMPS GPU engine is physically robust and numerically stable. The successful reproduction of Argon structures and LJ EoS using NHC thermostats provides a solid foundation for future relativistic QGP simulations.